ADAMA SCIENCE & TECHNOLOGY UNIVERSITY

CHAPTER 3

MOBILE DEVICE HARDWARE

The Mobile Computing Paradigm

What is the Mobile Computing Paradigm?

- Design philosophy balancing **performance vs. power vs. size**
- Optimizes for **efficiency per watt**, not maximum speed
- Operates within strict **battery, thermal, and space constraints**

TECHNICAL COMPARISON: SMARTPHONE VS. DESKTOP

DESKTOP COMPUTER

POWER BUDGET

65-300W
(UNLIMITED WALL POWER)

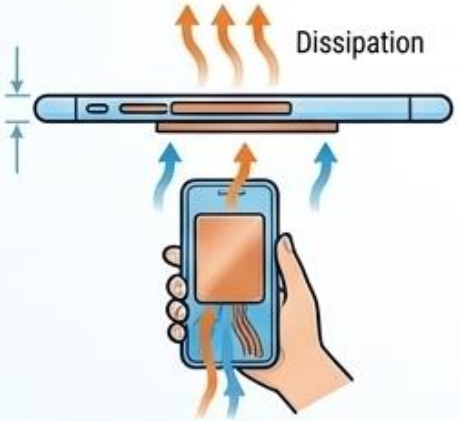
THERMAL DESIGN

EXPANDABLE TOWER, NO SIZE CONSTRAINT

SMARTPHONE

POWER BUDGET

3-10W
(BATTERY LIMITED)



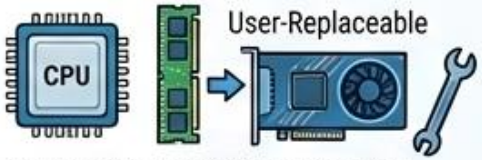
PASSIVE COOLING ONLY

DESKTOP COMPUTER

FORM FACTOR

EXPANDABLE TOWER, NO SIZE CONSTRAINT

UPGRADABILITY



REPLACE ANY COMPONENT

ALWAYS-ON

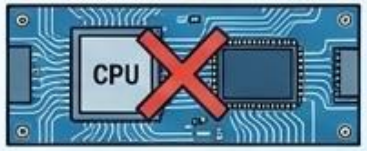
SLEEP MODE AVAILABLE

SMARTPHONE

FORM RACTOR

SEALED, 7-8MM THICK, 150-200G

UPGRADABILITY



SOLDERED, NO UPGRADES

ALWAYS CONNECTED, BACKGROUND LIMITS

The Mobile Computing Paradigm

The Efficiency Imperative

- Mobile CPUs: **10x better performance per watt** than desktop
- Achieved through:
  - ✓ ARM architecture,
  - ✓ heterogeneous cores,
  - ✓ specialized accelerators,
  - ✓ power gating

Key Insight: our phone isn't a weak computer; it's a specialized efficiency champion that does 90% of desktop tasks using 5% of the power.

System on Chip (SoC)

What is an SoC?

- **Single chip** integrating multiple processors
- Replaces separate chips (CPU, GPU, etc.) on a board
- **Benefits:** Smaller, faster, lower power (signals travel shorter distances)

COMPONENTS INSIDE A SYSTEM-ON-A-CHIP (SoC)

CPU (Central Processing Unit)

Function: General-purpose computation
Core processor that runs the operating system and most applications.

GPU (Graphics Processing Unit)

Function: Graphics rendering
Handles 3D graphics, user interface elements, and video decoding/encoding.

Modem (Modulator–Demodulator)

Function: Cellular connectivity.
Manages data connections with cellular networks, enabling wireless communication.

NPU (Neural Processing Unit)

Function: AI/ML acceleration.
Power-efficient processing of machine learning tasks like face recognition and image enhancement.

DSP (Digital Signal Processor)

Function: Audio/voice processing.
Handles compressed audio, voice recognition, and sound effects efficiently.

ISP (Image Signal Processor)

Function: Camera image processing.
Turns raw data from the camera sensor into high-quality photos and videos.

What is a Mobile Operating System?

Major SoC Vendors

Vendor	Series	Key Devices	Strength
Apple	A17 Pro, A18, M4	iPhone 15 Pro, iPad Pro	Best performance per watt
Qualcomm	Snapdragon 8 Gen 3	Samsung S24, OnePlus 12	AI performance, 5G modem
MediaTek	Dimensity 9300	Vivo X100, OPPO Find X7	Value flagship pricing
Samsung	Exynos 2400	Galaxy S24 (EU)	AMD GPU collaboration
Google	Tensor G4	Pixel 9	AI/ML optimization

CPU Architecture – Clock Speed

What is Clock Speed?

- **Clock Speed** – Measures the number of **CPU cycles completed per second**.
- Measured in **Hertz (Hz)**, commonly **Gigahertz (GHz)**.
- **1 GHz = 1 billion cycles per second**.
- Each clock cycle helps execute parts of the **instruction cycle** (fetch, decode, execute).
- **Higher clock speed = more operations per second**, but **not always faster** due to differences in CPU architecture, cores, cache, and efficiency.

CPU Architecture – Core Count

What are CPU Cores?

- **CPU Cores** – Independent processing units within a single processor chip.
- Allow **parallel execution** of multiple tasks simultaneously.
- More cores improve **multitasking and workload distribution**.
- Each core can run **its own instruction thread**.
- Modern mobile processors use **heterogeneous core designs** (performance cores + efficiency cores).

MODERN CPU HYBRID ARCHITECTURE

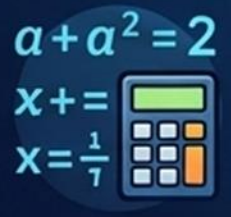
P-CORE VS E-CORE

PERFORMANCE CORE (P-Core)

- Designed for Maximum Speed
- High-Power Core
- Demanding Tasks

Gaming

Heavy Applications



Complex Calculations

Single-Thread Performance

EFFICIENCY CORE (E-Core)

- Optimized for Energy Efficiency
- Low-Power Core
- Background Tasks

Background Apps

Multitasking

Light Workloads

Costs

KEY BENEFITS

Improved Battery Life

Power Efficiency

Multitasking Performance

CPU Architecture – Cache Memory

What is CPU Cache?

- **CPU Cache** – Small, ultra-fast memory located on or near the CPU.
- Stores **frequently used data** to reduce access time to RAM.
- **Speeds up processing** by keeping data close to the CPU cores.
- Without cache, the CPU can be **idle most of the time (up to 98%)** waiting for data.
- Modern CPUs have **multiple cache levels**: L1 (fastest, smallest), L2 (larger, slower), L3 (shared, largest).

MEMORY HIERARCHY ORDER (FAST → SLOW)

SPEED DECREASING, LATENCY INCREASING, CAPACITY INCREASING

ULTRA-FAST LATENCY

SMALLEST CAPACITY

HIGHEST CAPACITY

L1 CACHE

CORE

Size: 64KB per core
Speed: 1-2 ns
Location: On-core

MEDIUM LATENCY

MEDIUM CAPACITY

L2 CACHE

Size: 4-8MB per cluster
Speed: 3-5 ns
Location: Near-core

L3 CACHE

Size: 8-16MB shared
Speed: 10-20 ns
Description: Shared by all cores

SLOW LATENCY

LARGE CAPACITY

RAM (SYSTEM MEMORY)

Size: 8-16GB
Speed: 50-100 ns
Location: Separate chip

HIGHEST LATENCY

HIGHEST CAPACITY

STORAGE (NVMe/SSD)

SSD

Size: 256GB-1TB
Speed: 10,000-100,000 ns
Description: Flash chips

CPU Architecture – Instruction Set Architecture (ISA)

What is ISA?

- **ISA (Instruction Set Architecture)** – Interface between **hardware and software**.
- Defines the **instructions a CPU can execute**.
- Impacts **CPU design, performance, and efficiency**.
- Two main types:
  - ✓ **CISC** – Complex instructions, fewer lines of code.
  - ✓ **RISC** – Simple instructions, fast and power-efficient.
- Modern CPUs may **combine both features**.

COMPARING PROCESSOR ARCHITECTURES: ARM vs. x86

x86 (CISC)		ARM (RISC)
MAXIMIZE WORK PER INSTRUCTION	DESIGN PHILOSOPHY	SIMPLIFY FOR SPEED AND EFFICIENCY
COMPLEX, MULTI-STEP	INSTRUCTION COMPLEXITY	SIMPLE, ATOMIC, SINGLE-STEP
65-300W	POWER CONSUMPTION	3-10W
ACTIVE (FANS) REQUIRED	COOLING	PASSIVE ONLY
DESKTOPS, SERVERS, LAPTOPS	PRIMARY MARKETS	MOBILE, EMBEDDED, IoT

WHY ARM DOMINATES MOBILE: SIMPLER INSTRUCTIONS = MORE EFFICIENT POWER USAGE, SMALLER CHIPS, EASIER THERMAL MANAGEMENT

GPU – Graphics Processing Unit

What is a GPU?

- **Specialized processor** for images, video, animations
- **Parallel architecture:** 1000+ simple cores vs. CPU's 6-8 complex cores
- Optimizes for **throughput** (total work), CPU optimizes for **latency** (single task speed)
- **CPU vs GPU**: CPU = single-task speed, GPU = many-task speed.

GPU – Graphics Processing Unit

Mobile GPU Landscape

GPU	Manufacturer	Key Features	Found In
Apple GPU	Apple	Tile-based rendering, Metal optimization	iPhone, iPad
Adreno 750	Qualcomm	Raw performance, ray tracing	Snapdragon 8 Gen 3
Immortalis-G720	ARM	Hardware ray tracing, VRS	Dimensity 9300
Xclipse 940	Samsung+AMD	RDNA3 architecture	Exynos 2400

GPU – Graphics Processing Unit

GPU Specifications

Spec	**Entry**	**Mid-Range**	**Flagship**
Clock speed	500 MHz	800 MHz	1.0-1.5 GHz
Execution units	2-4	6-8	10-16
Fill rate	2 GP/s	8 GP/s	20+ GP/s

NPU – Neural Processing Unit

What is an NPU?

- **Specialized hardware** for AI/ML workloads
- **10-100x faster** than CPU for neural networks
- **1/50th power** of cloud-based AI processing
- Enables **on-device privacy** (data never leaves phone)

NPU – Neural Processing Unit

NPU Performance (TOPS)

Chip	NPU	TOPS	Capability
Apple A18 Pro	Neural Engine	38	On-device LLM, image generation
Snapdragon 8 Gen 3	Hexagon	45	10B parameter models on device
Dimensity 9300	APU 790	35	Real-time video AI effects
Tensor G4	TPU	25	Call screening, Magic Eraser

TOPS = Tera Operations Per Second (trillions of AI operations)

NPU – Neural Processing Unit

Real-World AI Tasks

Task	Without NPU	With NPU	Benefit
Face recognition	500ms (CPU)	10ms	50x faster
Photo categorization	30s + cloud	1s on-device	Privacy, offline
Live translation	2s delay (cloud)	Real-time	No latency
Voice wake word	500mW power	10mW	50x efficiency

TOPS = Tera Operations Per Second (trillions of AI operations)

Memory Architecture – RAM

What is RAM?

- **Random Access Memory:** volatile, high-speed working memory
- **Volatile:** Data lost when power removed
- **Fast:** 50-100 nanosecond access time
- Stores **currently running apps** and OS data

Memory Architecture – RAM

Mobile RAM Technologies

Generation	Speed	Power Efficiency	Used In
LPDDR4X	4266 Mbps	Good	Budget to mid-range (2019-2022)
LPDDR5	6400 Mbps	Better	Flagship Android, iOS (2020-2023)
LPDDR5X	8533 Mbps	Best	Latest flagships (2023+)

LPDDR = Low Power Double Data Rate

Memory Architecture – RAM

OS Memory Management

Aspect	Android	iOS	HarmonyOS
Strategy	Keep many apps in RAM	Suspend to disk, resume fast	Predictive, adaptive
Overhead	+20-30% (Java/Kotlin)	Minimal (native)	Optimized
Effective efficiency	12GB Android ≈ 8GB iOS	Highly optimized	Can pool across devices

Memory Architecture – RAM

Organization Systems:

System	Approach	Real-World
Android	Keep everything out, piles everywhere	12 apps "open", system manages
iOS	Put away what not using, clean surface	4 apps active, others suspended
HarmonyOS	Borrow space from colleague's desk	Use RAM from tablet/watch/TV

Key Insight: RAM is your active workspace—too small and you're constantly filing/retrieving (slow), too large and you're wasting office space (power). The organization system matters as much as desk size.

Memory Architecture – Storage

What is Flash Storage?

- **Non-volatile memory:** Retains data without power
- Stores **OS, apps, photos, videos, documents**
- Slower than RAM, much faster than hard drives
- No moving parts = durable, compact

Memory Architecture – Storage

Storage Technology Comparison

Technology	Read Speed	Write Speed	Used In	Feel
NVMe (PCIe)	3,000 MB/s	2,500 MB/s	iPhone, iPad Pro	Instant
UFS 4.0	4,200 MB/s	2,800 MB/s	Flagship Android 2023+	Instant
UFS 3.1	2,100 MB/s	1,200 MB/s	Mid-range	Fast
UFS 2.2	1,000 MB/s	500 MB/s	Budget	Noticeable lag
eMMC 5.1	400 MB/s	200 MB/s	Very old	Frustrating

Memory Architecture – Storage

Real-World Impact

Operation	Fast (NVMe)	Slow (eMMC)	User Perception
App launch	1.2 seconds	4 seconds	"Snappy" vs "Slow phone"
Game level load	8 seconds	25 seconds	Engaged vs frustrated
Photo burst save	Instant	5 second wait	Captured vs missed
4K video edit	Smooth	Stuttering	Professional vs unusable

Memory Architecture – Storage

Real-World Impact

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Key Insight: Storage speed is like filing cabinet quality—modern systems let you grab any file instantly, while old systems have users staring at loading spinners.

Power Systems – Battery

What is a Lithium-Ion Battery?

- **Rechargeable cell** powering virtually all mobile devices
- **High energy density:** 250-300 Wh/L (more per volume than alternatives)
- **No memory effect:** Can charge at any level
- **Cycle life:** 500-800 full cycles to 80% capacity

Power Systems – Battery

Battery Specifications

Metric	Typical Range	Meaning
Capacity (mAh)	3,000-6,500	Total charge storage (higher = longer life)
Voltage	3.7V nominal, 4.2-4.5V full	Operating and peak voltage
Energy density	250-300 Wh/L	Power per unit volume
Cycle life	500-800 cycles	Full charge/discharge before degradation

Power Systems – Battery

Device Class Comparison

Category	Capacity	Usage	Charging Speed
Compact	3,000-3,800 mAh	1 day light use	20-30W
Mainstream	4,500-5,000 mAh	1 day moderate use	45-67W
Large/Ultra	5,000-5,500 mAh	1.5 days or heavy use	45-45W
Gaming	6,000-6,500 mAh	Heavy gaming	65-165W

Power Systems – Battery

Fast Charging

Standard	Max Power	Characteristics
USB PD 3.1	240W	Universal, negotiated
Qualcomm QC 5	100W+	Dual charge pumps
OPPO SuperVOOC	150-240W	Multi-cell battery
Apple Fast Charge	27W	Conservative, battery health focus

Key Insight: Battery is your fuel tank—size determines range, charging speed determines pump wait time. Fast charging generates heat, aging the tank faster.

Power Systems – PMIC & Thermal Management

What is a PMIC?

Power Management Integrated Circuit:

- **Electrical system manager** for mobile devices
- Converts battery voltage to levels required by components
- Controls charging speed, monitors temperature, ensures safety

Power Systems – PMIC & Thermal Management

PMIC Functions

Function	Description	Criticality
Voltage regulation	3.7V → 0.5V (CPU), 3.3V (RAM), etc.	Different components need different "fuel grades"
Charging control	Manages speed based on temperature	Prevents fires, extends battery life
Power distribution	Shuts off unused components	Saves every milliamp
Fuel gauging	Reports battery percentage	User trust
Safety monitoring	Stops if too hot/cold	Prevents damage, explosions

Power Systems – PMIC & Thermal Management

Thermal Management

State	Temperature	CPU Response	User Experience
Normal	<40°C	Full performance	Optimal
Warm	40-45°C	Slight throttling	Unnoticeable
Hot	45-50°C	Aggressive throttling	Frame drops, lag
Critical	>50°C	Emergency shutdown	App crashes

Thermal throttling: Reduces clock speeds to prevent damage, but reduces performance.

Key Insight: PMIC is like a smart gas station manager—right fuel, safe delivery, emergency stops. Thermal throttling is like an athlete slowing down to avoid collapse.

Sensors – Motion & Position

What are MEMS Sensors?

Micro-Electro-Mechanical Systems:

- **Microscopic machines** on silicon chips
- Detect physical forces (acceleration, rotation, pressure)
- Convert mechanical signals to electrical data

Sensors – Motion & Position

Accelerometer: Linear Motion

- Measures **acceleration** along X, Y, Z axes
- Detects: **gravity** (orientation), **movement** (shaking, steps), **vibration**

Spec	**Typical Value**
Axes	3 (X, Y, Z)
Range	±2g to ±16g
Resolution	16-bit
Sample rate	Up to 400 Hz

Sensors – Motion & Position

Gyroscope: Rotation

- Measures **angular velocity** (spin speed)
- Detects: **rotation** around pitch, roll, yaw axes
- Not affected by gravity (unlike accelerometer)

Combined: IMU (Inertial Measurement Unit)

- Accelerometer + Gyroscope + Magnetometer = 9-axis
- Enables: AR, VR, image stabilization, navigation without GPS

Sensors – Motion & Position

GPS/GNSS: Global Positioning

- **Satellite triangulation:** Measures signal travel time from multiple satellites
- Systems: GPS (US), GLONASS (Russia), Galileo (EU), BeiDou (China)

Metric	Typical Value
Accuracy	3-5 meters outdoors
Time to first fix (cold)	30-60 seconds
A-GPS assisted	2-5 seconds
Power consumption	50-150 mW

Sensors – Motion & Position

Real-World Gestures:

Gesture	Ball Movement	App Response
Shake	Rapid oscillations	Undo, refresh
Tilt	Presses one side	Screen rotation
Step	Periodic up/down	Pedometer
Free fall	No contact	Hard drive protection

Sensors – Motion & Position

Gyroscope: Spinning Top

Rotation	Axis	Use
Pitch (nod)	X	Flight simulators
Roll (tilt head)	Y	Racing games
Yaw (shake head)	Z	VR tracking

Key Insight: Sensors are like human senses—accelerometer is your inner ear (balance), gyroscope detects rotation, GPS is like asking multiple landmarks where you are.

Sensors – Environmental & Biometric

Environmental Sensors

Sensor	Measures	Range	Application
Ambient light	Brightness	1-100,000 lux	Auto-brightness, True Tone
Barometer	Air pressure	300-1100 hPa	Altitude, weather prediction
Temperature	Thermal state	Varies	Device protection, throttling
Proximity	Nearby objects	0-5 cm	Screen off during calls

Sensors – Environmental & Biometric

Barometer applications:

- **Altitude:** 1 hPa ≈ 8 meters elevation change
- **Weather:** Falling pressure = storm approaching
- **GPS:** Faster altitude lock
- **Indoors:** Floor-level detection

Sensors – Environmental & Biometric

Biometric Sensors

Fingerprint types:

Type	Technology	Security	Speed
Capacitive	Electrical field maps ridges	High (1/50k)	<100ms
Optical	Camera image	Medium (spoofable)	200-300ms
Ultrasonic	3D sound wave mapping	Very high (1/100k)	200-300ms

Sensors – Environmental & Biometric

Biometric Sensors

Face recognition generations:

Gen	Technology	Security	Fooled By
1st (2017)	2D camera	Low	Photo, video
2nd (2019)	IR + depth	Medium	High-res mask
3rd (2021)	30,000 dot projector	High	Elaborate 3D mask
4th (2024)	Multi-spectrum + liveness	Very high	Nothing practical

Key Insight: Biometric sensors are like security guards—each generation gets better at detecting fakes. Environmental sensors are like weather stations, giving your phone awareness of its surroundings.

Display & Touch Technologies

What is a Display Panel?

- **Visual output device** presenting information via controlled light
- Must balance: image quality, power consumption, durability, thickness

Display & Touch Technologies

Display Technology Comparison

Technology	Mechanism	Strengths	Weaknesses
OLED	Self-emitting pixels	Perfect blacks, thin, flexible	Burn-in, expensive, PWM sensitivity
AMOLED	Active matrix OLED	Same as OLED, more efficient	Same as OLED
LTPO OLED	Variable refresh transistors	1-120Hz adaptive	Complex, costly
Mini-LED	Thousands of LED zones	Near-OLED, no burn-in	Thicker, blooming
LCD	Backlight + liquid crystals	Cheap, bright, reliable	Gray blacks, thicker

Display & Touch Technologies

Key Display Specifications

Spec	Good	Better	Best
Resolution	FHD (1080p)	QHD (1440p)	4K
Refresh rate	60Hz	90/120Hz	144Hz+
Brightness	500 nits	1000 nits	2000+ nits
HDR	HDR10	HDR10+	Dolby Vision

Display & Touch Technologies

Touchscreen Technology

Haptic Feedback

Type	Mechanism	Feel	Used In
ERM	Eccentric rotating mass	Buzzy, imprecise	Budget phones
LRA	Linear resonant actuator	Sharp, precise	iPhone, flagships

Display & Touch Technologies

Display Technologies: Canvas Types

Display	Canvas	Characteristics
OLED	Self-illuminating	Each pixel = tiny light bulb. Perfect black when off, vibrant, but burn-in if static image
LCD	Backlit canvas	Always backlit, can't show true black (gray), but reliable and cheap
LTPO OLED	Smart canvas	Auto-adjusts refresh—fast for action, slow for static, saves energy

Display & Touch Technologies

Refresh Rate: Flipbook Speed

Rate	Speed	Experience	Battery
60Hz	Standard	Acceptable	Baseline
90Hz	Smooth	Noticeably better	+10%
120Hz	Very smooth	Butter scrolling	+20%
Adaptive	Smart	Matches content	Optimized

Key Insight: Display is your window to the digital world—OLED is like a smart window adjusting transparency, touch is the handle responding instantly to your grip.

Camera System Architecture

What is a Computational Camera?

- Combines **optical hardware** (lenses, sensors) with **software processing** (ISP, NPU)
- Modern mobile photography: **computation > pure optics**

Camera System Architecture

Multi-Camera Array

Camera	Purpose	Typical Specs	Use Case
Ultra-wide	Landscape, architecture	12MP, f/2.2, 120°	Wide scenes, tight spaces
Main/Wide	Standard photography	48-200MP, f/1.4-1.8	Everyday photos
Telephoto	Optical zoom, portraits	12MP, 3x-10x optical	Distant subjects
Macro	Extreme close-up	5MP, very close focus	Details, textures
Depth/ToF	Distance mapping	Low-res, IR	Portrait blur, AR
Front	Selfies, video calls	12-32MP, wide angle	Self-portraits

Camera System Architecture

Key Camera Specifications

Spec	What It Means	Reality Check
Megapixels	Sensor resolution	12MP sufficient; bigger pixels > more pixels
Sensor size	Physical area	Larger = more light = better quality
Pixel size	Micrometers (µm)	1.4µm = excellent low light; 0.7µm = noisy
Aperture	Light opening (f-number)	f/1.4 = lots of light; f/2.8 = less light
OIS	Optical stabilization	Physical lens movement; 2-3 stops improvement
EIS	Electronic stabilization	Software crop; reduces resolution

Key Insight: Modern phone cameras are like having a professional photography team in your pocket—each lens is a specialist, and AI is the darkroom technician developing perfect photos instantly.

Connectivity Architecture

What is Cellular Communication?

- Wireless data/voice transmission via **network of base stations** (cell towers)
- Each generation (G) = major leap in speed, latency, capabilities

Connectivity Architecture

Cellular Evolution

Gen	Era	Speed	Latency	Key Capability
1G	1980s	-	-	Analog voice
2G	1990s	0.1 Mbps	600ms	SMS, basic data
3G	2000s	2 Mbps	100ms	Mobile web, video calls
4G LTE	2010s	100 Mbps	50ms	HD streaming, apps
5G Sub-6	2020s	1 Gbps	20ms	4K streaming, cloud gaming
5G mmWave	2024+	10 Gbps	1ms	VR/AR, instant response

Connectivity Architecture

Wi-Fi Evolution

Standard	Year	Speed	Key Advance
Wi-Fi 5 (ac)	2014	3.5 Gbps	5 GHz focus, beamforming
Wi-Fi 6 (ax)	2019	9.6 Gbps	OFDMA, crowded area efficiency
Wi-Fi 6E	2021	9.6 Gbps	6 GHz band, less interference
Wi-Fi 7 (be)	2024	46 Gbps	320 MHz channels, MLO

Connectivity Architecture

Other Connectivity

Tech	Range	Speed	Primary Use
Bluetooth 5.3	100m	2 Mbps	Audio, wearables
NFC	<10 cm	424 kbps	Payments, transit, tap-to-pair
UWB	10-200m	Low data	Precision location, AirTag, car keys
Thread	Mesh network	Low data	Smart home, self-healing networks

Key Insight: Connectivity is like transportation infrastructure—5G is the highway system (fast, long-distance), Wi-Fi is your driveway (private, local), NFC is a secret handshake (very close, very secure).

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END *of* CHAPTER 3